

Lecture 4 Team Projects

Benchmark Spreads and Term Structure

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1 Project 1: Build and Read a Government Yield Curve

Scenario. Prepare a desk briefing comparing yield curves on two materially different dates.

Objectives. Construct comparable curves; measure level, slope, and curvature changes; connect movements cautiously to policy and macro news.

Sources. FRED, U.S. Treasury, or Bank of Canada yields; Federal Reserve or Bank of Canada releases.

Deliverables. Two curve plots, 2s10s and 5s30s slopes, curvature measure, event timeline, desk conclusion, and a 10–12 minute presentation.

Rubric (100 points). Data and curves 25; measures 25; event interpretation 20; risk implications 20; presentation and attribution 10.

2 Project 2: Bootstrap a Discount Curve by Hand

Scenario. Derive discount factors and spot rates from a small set of coupon-bearing government securities.

Objectives. Explain sequential bootstrapping; use four to six instruments; validate through repricing; assess interpolation risk.

Sources. Instructor-consistent price snapshot or official U.S. Treasury/Bank of Canada terms with documented same-date prices.

Deliverables. Cash-flow matrix, bootstrap calculations, discount and spot curves, repricing errors, convention note, and a 10–12 minute presentation.

Rubric (100 points). Setup 20; bootstrap accuracy 35; validation 20; limitations 15; presentation and reproducibility 10.